"Most frequent waiting queries" and "Queries that waited the most" that can be helpful for investigations.

Setting Statement Timeout

In this section you'll follow a similar pattern to how you set a lock_timeout but focus on the statement_timeout.

Note that queries getting canceled for taking too long cause an application error and a bad experience, especially when they're related to a user-facing Rails application query compared with a non-user facing background job query. In both cases the associated queries need to be fixed so that they don't exceed the timeout. The solutions will depend highly on the circumstances.

Setting too low of a value for statement_timeout cancels queries unnecessarily. Not leaving it set at all allows queries to run forever and consume resources that are finite, such as a database connection. A happy middle ground is to use the statement_timeout and select a value that balances these two extremes.

In a Ruby on Rails web application, user-facing queries should run in less than 250ms for ecommerce or consumer web sites and applications. In B2B Enterprise SaaS applications, there may be less appetite for query optimization, or tolerance for slower queries. However, fast queries are a good goal regardless of the type of application, because in the long run fast queries make it easier for the same server to achieve higher concurrency, and for your team to avoid costs related to scaling up.

If your application does not have a statement timeout set at all, try starting with a very high one like 10 minutes or 5 minutes. As you fix queries and some time passes without observing cancellations from timeouts, lower the timeout to 5 minutes or 1 minute. You may want to allocate some database connections within your limited set for especially slow background queries. You may also want to allow a higher timeout value for background job queries that aren't user-facing.

Lock Timeout and Statement Timeout are two options worth setting.

In the next section you'll consider how Active Record caches your schema.